Performance Testing

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Custom Python load script `tools/perf_load.py` (httpx + threads, psutil for CPU/RAM sampling)
Type of Testing	Load testing (1, 10 and 25 concurrent virtual users, 20 s each)
Target Module / API	`POST /home-budget`, `POST /party-budget`, `POST /jewelry-budget`, `GET /recommendation-history` (rotating mix), after register + login
Test Environment	Local: Linux, 2 vCPU, 8 GB RAM, single Uvicorn worker, load generator on the same machine. **Offline-fallback mode (no `GOOGLE_API_KEY`)**, so results measure PocketSmart's own overhead (validation, budget engine, SQLite) and exclude Gemini network latency
Test Date	03 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline: one user cycling through all four endpoints	1	20	Fast responses, 0 errors
2	Typical classroom load	10	20	Avg < 2 s, 0 errors
3	Peak load	25	20	Avg < 2 s, max < 5 s, error rate < 1 %
4	Functional regression under test (pytest)	1	18	43 of 43 tests pass

Step 3: Performance Test Results (25 virtual users)

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.104 s	Pass	0.006 s at 1 user, 0.048 s at 10 users
2	Response Time (Max)	< 5 seconds	2.055 s	Pass	p95 = 0.387 s; maximum most likely caused by bcrypt login and registration bursts at start-up
3	Throughput (Req/sec)	Not specified	135 req/s	Pass	159 req/s at 1 user, 172 req/s at 10 users
4	Error Rate	< 1%	0.00 %	Pass	0 failures in 2,724 requests (3,178 and 3,467 requests in the 1 and 10 user runs)

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
5	CPU Utilization	< 80%	avg 65 % of the 2-core machine (130 % of one core); peak 99 %	Partial	Average passes; peak is high because the load generator shares the same 2 cores
6	Memory Utilization	< 80%	91 MB peak (about 1 % of 8 GB)	Pass	Server RSS grew only 74 MB to 91 MB across the runs

Step 4: Observations & Analysis

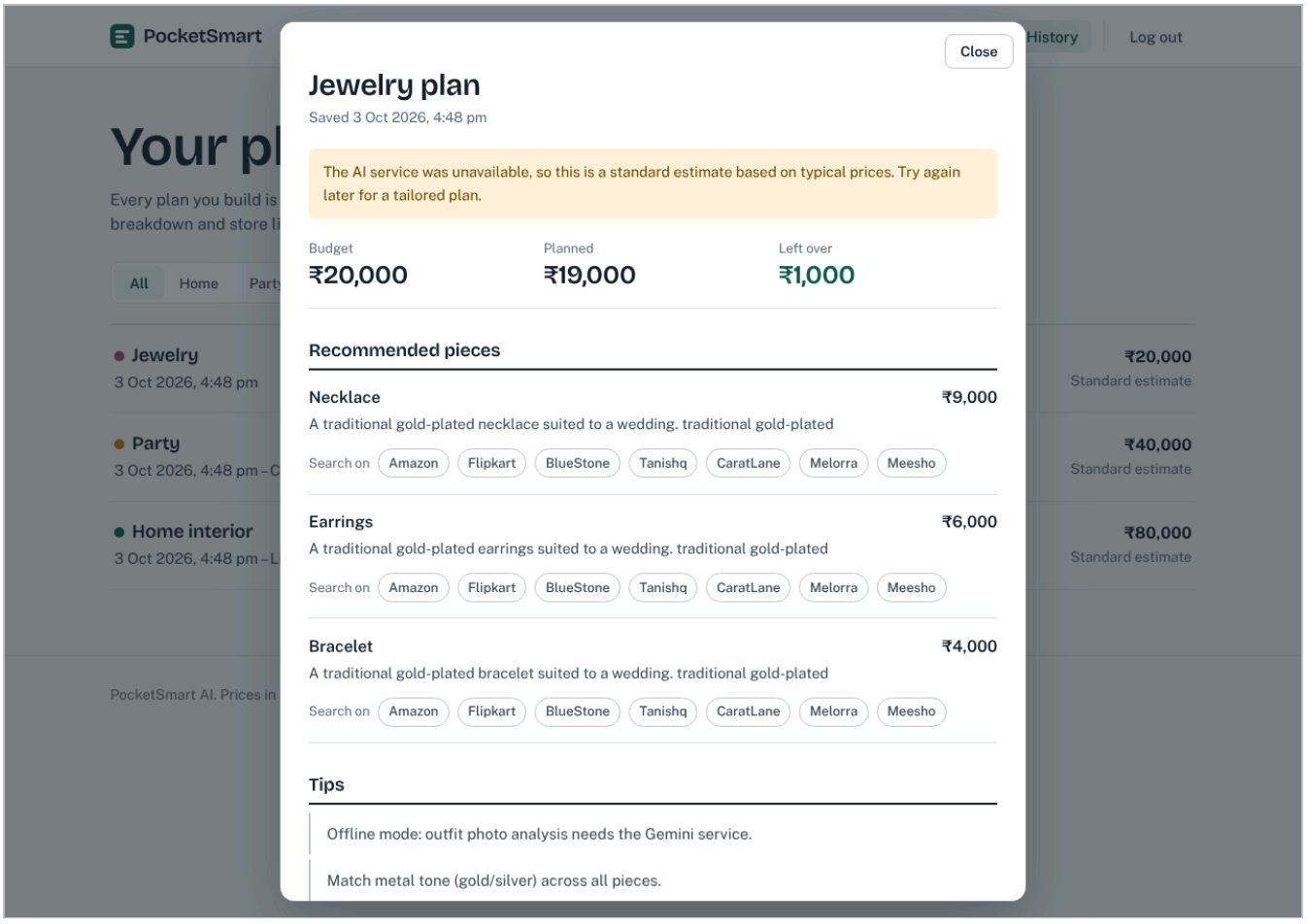
- PocketSmart's own processing is light: average latency stays at about 0.1 s even with 25 simultaneous users, and no request failed.
- In production the response time is dominated by the Gemini call (typically a few seconds). Blocking AI calls run in FastAPI's threadpool, so a slow Gemini reply for one user does not block others.
- Latency tails (max about 2 s) most likely come from bcrypt hashing at registration and login, which is intentionally slow for security.
- **Not measured:** live Gemini latency, because no API key was available during automated testing. The team should run `python tools/check_gemini.py` and time a planner request with the real key before the demo and add the figure here.
- Scalability next steps: multiple Uvicorn workers, PostgreSQL, a shared session store (Redis) and response caching.

Step 5: Screenshots / Evidence

Raw outputs are stored in `6.Project Testing/evidence/` (`load_1_users.txt`, `load_10_users.txt`, `load_25_users.txt`, `pytest_results.txt`).

```
virtual users      : 25
duration (s)       : 20.2
requests           : 2724
throughput (req/s) : 135.0
avg response (s)   : 0.104
p95 response (s)   : 0.387
max response (s)   : 2.055
error rate (%)     : 0.00
server CPU avg/max : 130% / 198% (of one core)
server RSS avg/max : 86 MB / 91 MB
```

Functional test run: `43 passed` (auth flow, validation edge cases, upload rejection, budget adherence, history privacy, Gemini success / failure / model fallback with mocks).


PocketSmart
History
Log out
Your pl
Every plan you build is
breakdown and store li
All
Home
Party
Jewelry
3 Oct 2026, 4:48 pm
₹20,000
Standard estimate
Party
3 Oct 2026, 4:48 pm – C
₹40,000
Standard estimate
Home interior
3 Oct 2026, 4:48 pm – L
₹80,000
Standard estimate
PocketSmart AI. Prices in
Close
Jewelry plan
Saved 3 Oct 2026, 4:48 pm
The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.
Budget
₹20,000
Planned
₹19,000
Left over
₹1,000
Recommended pieces
Necklace
₹9,000
A traditional gold-plated necklace suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Earrings
₹6,000
A traditional gold-plated earrings suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Bracelet
₹4,000
A traditional gold-plated bracelet suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Tips
Offline mode: outfit photo analysis needs the Gemini service.
Match metal tone (gold/silver) across all pieces.